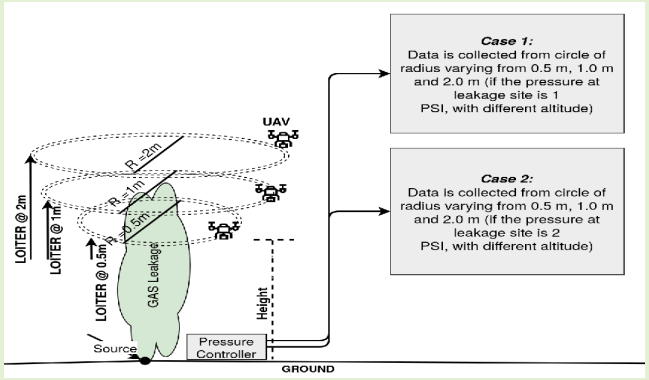
Detection and Estimation of Natural Gas Leakage Using UAV by Machine Learning Algorithms

Sarvesh Sonkar, Prashant Kumar, Riya Catherine George, Deepu Philip, and Ajoy Kanti Ghosh

***Abstract*—Presently, unmanned aerial vehicles (UAVs) have a broad spectrum of applications, in which regular monitoring of critical infrastructure assets is an important risk mitigation usage. This paper presents a novel approach for modeling leakage from natural gas pipelines using machine learning algorithms operating on the gas leakage detection (Methane) data collected using an UAV having a gas sensor and a light detecting and ranging (LiDAR) payload. The detection system has a lightweight design along with a small form factor to ensure compatibility with most autonomous mobile platforms like UAVs or wheeled robots. Two experiments were conducted to collect the gas leakage detection data at various loitering heights in real atmospheric conditions before applying the estimation algorithms. The first experiment measured natural gas leakage data in varying loitering radiuses and altitudes for leakage pressures around one pound per square inch (PSI), while the second experiment did the same for leakage pressures around two PSI. Real atmospheric conditions were incorporated along with additional aspects of propeller down-wash and environmental air movements that can influence leakage detection. Two estimation algorithms, viz., reduced support vector machine (RSVM) and artificial neural network (ANN) were applied to the leakage data collected from the UAV experiments. It was found that the ANN approach resulted in more faster and accurate detection than RSVM that heavily depended on the kernel function for its performance. The efficacy of leakage detection of natural gas using a UAV payload was demonstrated, which is a faster and cost-effective alternative to the manual inspection process.**



***Index Terms*— ANN, SVM, gas-leak, cross-wind air speed (2–4 m/s), Figaro (TGS2611), LiDAR, microSD card, regression, sensor, UAV.**

I. Introduction

UNMANNED aerial vehicles (UAV) or drones are becoming popular for scientific studies [1], especially in dispersed data gathering and remote sensing [2] in the defence and industrial sectors. Autonomous mobile platforms with embedded systems can gather environmental samples with greater efficacy than a human operator due to their higher spatial-temporal resolution. Increased pollution is a major cause of poor urban air quality, which is hazardous to public health. Many respiratory and heart diseases [3] triggered by environmental issues can be linked to gas emissions [4], [5]. Exhaust gases [6] from vehicles, industries, fires, chemical mishaps, etc. can discharge pollutants into the atmosphere. Due to the colorless and tasteless properties [7], [8] of volatile organic compounds, it becomes difficult for humans to detect leaks early and take preventative actions.

Stringent regulations necessitate better technical advancements to prevent air pollution from reaching dangerous level [9], along with minimizing property damage while ensuring worker and public safety. Many monitoring systems utilize a static network of sensors [10], [11], which is insufficient to provide the spatially dispersed concentration measurements required to obtain true estimates of gas dispersion [12], [13] along with pinpointing the location of the leak. The direct

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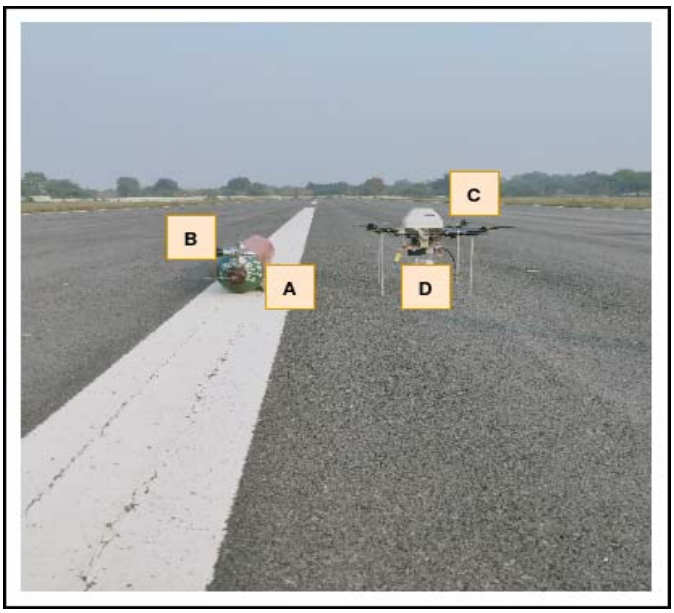


Fig. 1. Experimental setup with UAV and Methane cylinder.

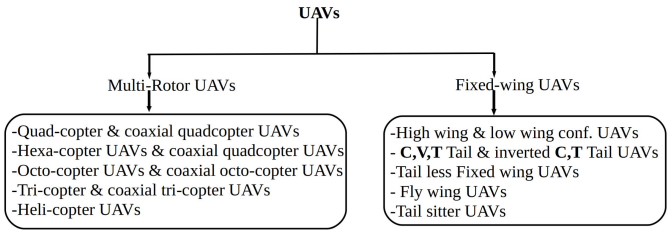


Fig. 2. Types of UAVs.

interaction between the chemical substances and the sensor surface tends to influence its response; the low-cost resistivity-based gas sensors tend to provide erroneous estimates. Thus, both the cost and deployment issues make the stationary sensor network a lesser lucrative option. Mobile platforms can play a major role in autonomous monitoring by offering various technologies that are capable to adapt to different scenarios.

This research presents an electronic platform of sensors that can work as a chemical sensing system for diverse environmental monitoring [14], like gas leak detection, pollution mapping, etc. Figure(1) depicts the UAV based system for gas leak detection, where "A" is the methane cylinder, "B" is the analog pressure sensor to measure the outlet pressure, "C" is the quad-copter UAV working as the mobility platform, and "D" is the gas-sensing payload. The system is designed using an eight-bit microcontroller unit (MCU) known as an Arduino board to control the payload control and log the data. Additionally, a digitally controlled analog interface exists to manage the "Figaro (TGS2611)" gas sensors along with the local recording of gas concentration in PPM and UAV height in a micro-SD card, resulting in the leakage dataset (gas concentration in PPM at various height). "TGS2611" has inbuilt filter material that eliminates the influence of interference gases thereby improving the sensitivity of detection for methane in natural gas, making it suitable for natural gas leakage detection. Finally, developers may utilize the IEEE 802.15.4-compliant short-range 2.4-GHz radio and the GPS interface [14] to create apps for real-time data exchange and/or alerts.

The electronic payload weighs less than 150 grams, requiring a power supply of 5volt [15], having dimensions of 4cm by 6cm. The lightweight design ensures compatibility with diverse mobility platforms, including UAVs. UAV's mobility allows for real-time monitoring and surveillance [16], [14] of air quality while the system is in motion. UAV's energy autonomy that accounts for the system's weight (payload) and electrical power consumption is the fundamental design criteria to locate and map gas leaks. Presently, no commercial multi-copter solutions exist that can deliver more than one hour of endurance on a single charge while carrying gas (TGS2611) and LiDAR sensor. Unlike existing research, we focused on enhancing the complete system including the payload, carrier, and energy autonomy of the sensor system. Our modular sensors architecture allows for the application of machine learning (ML) algorithms [17] to estimate leakage concentrations in different scenarios. Using these estimation techniques [18], we could determine the ideal settings for multi-copter speed, energy consumption, gas localization, and detection probability. Compared with existing methods presented in [11] and [19] that utilizes the remote methane leak detector (RMLD) sensor and tunable laser diode absorption spectrometer based methods [20] to detect natural gas leakage, the proposed approach using UAVs is modular and cost-effective.

The rest of this paper is organized into three sections as follows. Section II discusses the relevant background of the sensor-based payload, while Section III contains the proposed system architecture that elaborates on the payload modeling and implementation of the estimation techniques. Experimental results along with a comparison of alternatives are provided in Section IV . The last section of this paper concludes by summarizing key findings and eliciting contributions to existing research.

II. Background

Unmanned aerial vehicles (UAVs), popularly known as Drones are classified into two viz., (i) multi-rotor, and (ii) fixed-wing, based on their configuration [21] as illustrated in Figure(2). Multi-rotor UAVs have multiple propellers resulting in configurations like quadcopter, hexacopter [22], [23], octacopter [24], etc. When two propellers are assembled in a co-axial manner [25] causing single directional resultant thrust creates either a coaxial quadcopter, hexacopter, or octacopter [26]. The fixed-wing UAVs follow the traditional aircraft design that is quite efficient [27], [28], especially in endurance [29]. Typical mission requirements make low altitude long endurance (LALE) [30], [31] and high altitude long endurance (HALE) [32], [33] fixed-wing UAVs very important. Based on the mounting of the thruster for forward motion, fixed-wing UAVs can be either a puller or pusher [30]. Additionally, the tail-sitter UAVs are also fixed-wing UAVs that can operate without a runway [34], achieving vertical take-off and landing (VTOL).

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UAVs have matured beyond their military origins into civilian applications like essential data gathering and remote sensing due to two of their inherent benefits of high resolution and positional precision. Other applications include earth system knowledge [35], environmental science research, locating and mapping geographical regions for 3-D imagery, monitoring emissions, pinpointing the site of a gas leak [36], etc. Recently, UAVs are also tested for indoor emergency response. Most civilian systems suffer from two major flaws viz., (i) limited flying autonomy, and (ii) large size-to-payload ratio. Limited reported research exists that focus on simultaneously optimizing flight route and energy consumption of chemical sensors.

Some researchers have proposed using solar-powered UAVs for remote sensing and wide-area monitoring [37], [38] to overcome limited flying autonomy constraint. However, such systems are based on a fixed-wing architecture with large surface area (wing and fuselage) to facilitate the mounting of photovoltaic (PV) panels, thereby resulting in additional limitations like imprecise positioning, inability to hover, and always requiring sunlight [39].

Two primary technologies are utilized for autonomous distributed monitoring of hydrocarbon gases (Compressed Natural Gas (CNG), Liquified Petroleum Gas (LPG)) viz., (i) visual-based, and (ii) sensor (resistive and electrochemical) based [40], [41]. Sensor-based payloads have the advantages of lower weight, power consumption, and cost in comparison to the visual-based gas-optical systems. Studies have shown that chemical resistance technology (MOX sensors) [10], [42] provide rapid response and maintenance free long life while maintaining low power consumption. Other low-power alternatives (duty-cycle based methods) are demonstrated in research to circumvent the MOXs' power consumption [43] restriction, thereby enabling a longer life cycle for battery-powered embedded electronic devices.

Recently, the aerial robotics field started incorporating machine learning models for various applications. Major applications include controller design, parameter estimation for aircraft design, and payload processing like images stitching, object detection, and recognition [44]. Artificial Neural Networks (ANN) [11], [13], and Support Vector Machine (SVM) [45], [46] are two popular models used for estimation and detection applications.

III. System Architecture

This section broadly describes the VTOL system architecture, with two dedicated subsections to describe payload modeling and the implementation of machine learning algorithms for gas-leak detection. Our VTOL UAV (also known as multicopter UAV) has four propellers powered by four brush-less DC motors providing the thrust. Typical VTOL UAV system has five sub-systems viz., (i) airframe, (ii) propulsion, (iii) control, (iv) communications, and (v) payload. The mission requirements and payload capacity determine the design of the airframe and propulsion system. We took the mission requirement as flying at a low altitude for 40 minutes (endurance) while carrying both gas-leak sensor and LiDAR sensor as payloads. Figure (3) depicts the block diagram of the VTOL UAV with its payload system.

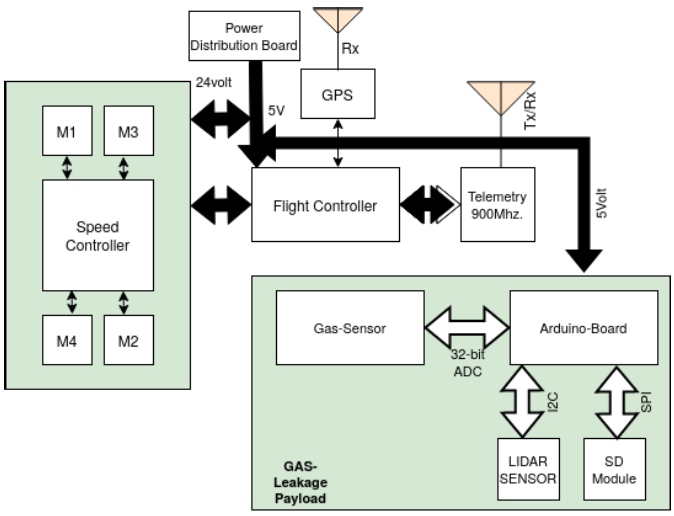


Fig. 3. Control block diagram with payload.

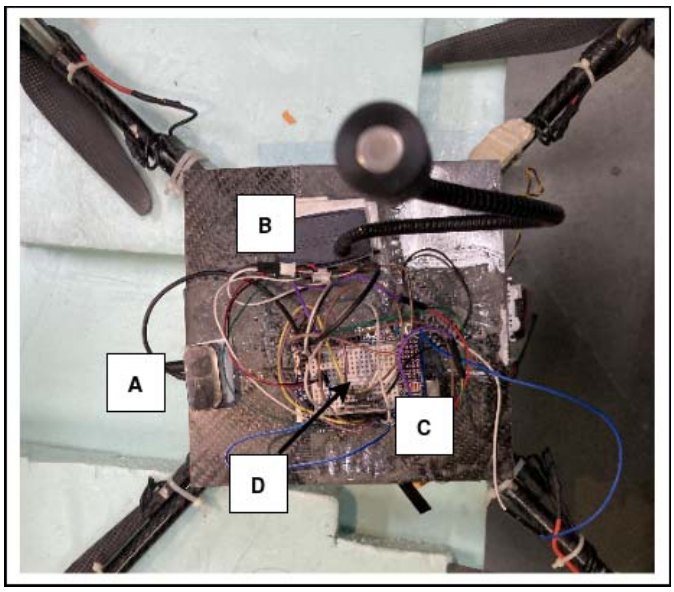


Fig. 4. Embedded payload design and acquisition system.

The training and testing of both algorithms were performed on intel (i5) 7th generation, 2.4GHz processor with 8GB memory.

A. Payload Modeling

The main focus of this research is on the payload of the VTOL UAV, whose schematic is shown in Figure (3) along with the real embedded hardware platform with micro-controller unit is depicted in Figure (4). The design of the mobile gas sensing payload has four major hardware components viz., (A) light detection and ranging (LiDAR) sensor, (B) gas leak detector, (C) micro-controller board, and (D) SD card in addition to the 5volt power supply. Figure (4) utilises the nomenclature (A, B, C, D) and the payload dimensions are 15×10cms with overall weight around 150grams. An eight-bit micro-controller based board having the ATmega328 MCU chip from Arduino company is used, which has ADC, I2C, SPI, PWM, and UART pins for interface

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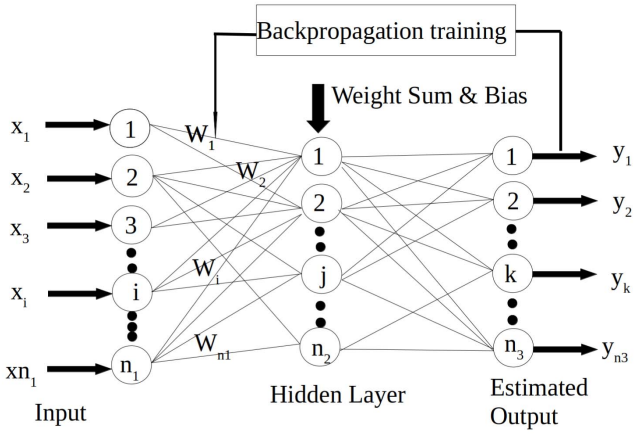


Fig. 5. Neural network layered architecture.

with various sensors. The serial peripheral interface (SPI) is a high-speed full-duplex communication protocol between master and slave, where MCU is the master and SD card is the slave. Similarly, the inter-integrated circuit (I2C) is a four-wire protocol including power and ground wire connection, while the analog to digital converter (ADC) connection is used to interface gas sensor, while $v_{ref}(3.3/5V)$ voltage pins are used for both sensors. Figure (3) illustrates the connection block diagram between two sensors (LiDAR & Gas-leak) and an SD card. Sensor fusion protocols listed below are used to combine sensors.

- LiDAR sensor is connected with MCU using **I2C protocol**
- SD card connected via **SPI protocols**
- Gas-leakage sensor connected via **ADC protocols**

B. Implementation

1) Neural Network Regression: ANNs are among the oldest and popular artificial intelligence (AI) tools, which mimics neural computing to solve unstructured problems that cannot be handled using mathematical formulas. ANN facilitates estimations for both linear and non-linear models without specifying a precise functional form. Also, ANN-based regressions are frequently used to model scenarios like rainfall-runoff, stream-flow, weather forecasting, disaster management, groundwater management, etc.

This research combines a feed-forward neural network with an error back-propagation (BP) [18] training technique, known as the BP neural network. As depicted in Figure (5) the BP neural network comprises of one input layer, one or more hidden layers, and one output layer. The number of hidden layers, number of hidden neurons (nodes), and transfer function are critical elements in constructing the topological structure of the BP network. The Kolmogorov theorem established that a single hidden layer can approximate any sophisticated nonlinear function and further established a nonlinear mapping between the input and output layers for an ANN. Hence, this research uses a three-layer BP neural network and illustrates the topological setup and activation function selection.

The pseudocode for the neural network regression is provided in Algorithm (1), where the feed-forward and back-propagation are summarised. Here, $(x^{(i)}, w)$ is the leakage data set (input) in part per million(PPM) with its corresponding weights, (l, E_{ij}^{l}) is the number of layers and error generated in each layer, and $(y^{(i)})$ is the ANN output(height in meters). The estimated model was tested using the original gas-leakage data set, which is detailed in the Experiments & Results section.

Algorithm 1 The BP Neural Network Pseudocode

procedure TRAIN(Leakage Data-Set)
  x ← Leakage data set (p×q) ▷ **i/p size & q=1**
  y ← Output
  w ← Weights of layers
  l ← No. of layers in network, 1…L
  $\{E_{ij}^{l}\}$ ← Error of Node j in layer l
  $\{t_{ij}^{l}\}$ ← 0 for all l, i, j ▷ **Total Error of Network**
  for $i = 1$ to p **do**
    $a^{l} \leftarrow (x^{(i)}, w)$ ▷ **feed-forward Step**
    $e^{l} \leftarrow a(L) - y(i)$ ▷ **Error of each layers**
    $\{t_{ij}^{l}\} \leftarrow \{t_{ij}^{l}\} + a_{i}^{l}.t_{i}^{l+1}$
    if $j \neq 0$ **then**
      $\{E_{ij}^{l}\} \leftarrow \frac{1}{p}.\{t_{ij}^{l}\} + \lambda\{w_{ij}^{l}\}$
    else
      $\{E_{ij}^{l}\} \leftarrow \frac{1}{p}.\{t_{ij}^{l}\}$
      $\{w_{ij}^{l}\} = N$
      where $\frac{\partial J(w)}{\partial N} = \{E_{ij}^{l}\}$
    end if
  end for
end procedure

2) Support Vector Machine: The gas-leakage data-set is $\{(x_i, y_i), i = 1, \ldots.., n\}$, where the input sequence $x_i \in R^d$ and the output of the regression is $y_i \in R$. The relation between input (x_i) and output target data (y_i) is shown in Equations (1), where w is the coefficient vector of regression $(w \in R^d)$, $\phi(x_i)$ is the transformation function, b is the bias function and R is the intercept.

$$y_i = w.\phi(x_i) + b \tag{1}$$

$$k(x_i, y_i) = (x_i^t, y_i + c)^n \tag{2}$$

Equations (1) and (2) are also known as regression equations. Also, in Equation (2) $c = 0$ is the kernel, also known as the homogeneous kernel, n is the order of the polynomial, and t is the structural parameter. The pseudo code for the support vector regression used here, popularly known as Reduced SVM (RSVM) [18], [45] is summarized in Algorithm (2).

IV. Experiments and Results

This section details our experiments and their results in two subsections. The first subsection discusses all case studies and experiments, while the next subsection provides estimations and detection results. As mentioned earlier, two different ML algorithms, viz., (i) ANN, and (ii) RSVM were used for estimating the natural gas leakage. The results subsection

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Algorithm 2 Support Vector Regression

Require: x_i, y_i, b, K ▷ Training Data-set(gas-leak)
Ensure: (1)$k(x_i, y_i) = (x_i^t, y_i + c)^n$
Ensure: (2) $y_i = w.\phi(x_i) + b$
1: [U,D] = Eigen Decomposition
2: $\tilde{K} = USDU^T$ with S = sign(D)
3: $[\tilde{\alpha}, b]$ = SvmSlover$(x_i, y_i, \tilde{K}, b)$
4: $\alpha = USU^T\tilde{\alpha}$
5: return α, b

compares the performance of both ML algorithms in real-time operational conditions.

A. Experiments

As discussed in section (III), we designed and prototyped the multi-copter with the payload of GAS & LiDAR sensors. Two scenarios based on the leakage pressures, referred as cases, were studied using three real-time experiments for each case with the multi-copter UAV to generate the gas-leakage detection dataset required to experimentally model these cases. The multi-copter UAV was operated in the **"Loiter"** maneuver for all three experiments, where the UAV circles around the designated loiter point in a constant radius recording the gas concentrations at regular intervals. The radius of the loiter circle becomes a control parameter for each experiment, which is limited by the UAV's bank-threshold angle.

The loitering height of the UAV, denoted by h, was varied from one meter upwards continuously, where $h \in R$. As the height of the UAV increases, it moves higher from the leakage source, thereby providing more space for the gas to dissipate causing a reduction in gas concentrations. Such concentration reductions can influence the detection efficiencies of the UAV payload.

The loiter circle radius was set at 0.5m for the first experiment, while the loiter radius for the second and third experiments were of one meter and two meters respectively. The number of values recorded per loiter circle differs according to its radius, but we ensured that a minimum of two values is recorded per loiter. As the loiter circle radius increased, the UAV circled farther from the source of leakage due to increased perimeter, thereby capturing more values per loiter. Hence, each experiment for a particular case consisted of n loiters, where n varied from one experiment to another.

Also, in both cases, realistic windy atmospheric conditions were included. The presence of crosswind allows for faster dissipation of the leaking gas, thereby significantly reducing its concentrations. So to model the effect of crosswind on the efficacy of the GAS sensor, wind speeds of 2∼4m/s were measured using handheld anemometer and recorded for all experiments. Due to the presence of crosswind, it is almost impossible to obtain a constant value of gas leakage detection per loiter circle. We summarize both cases in the following subsections.

1) Case-1: The first case was designed to study minor gas leaks, where the pressure of gas-leakage is ∼1 pound per square inch (PSI). The unit of leakage-detection data is parts per million (PPM), which is recorded along with corresponding height (h) of the UAV at regular intervals per loiter circle. This resulted in the leakage detection datasets for this case.

2) Case-2: The gas-leakage pressure in the second case was at ∼2 PSI, thereby simulating moderate gas leaks. This case also maintained the UAV loiter circle at 0.5m, one meter, and two meters for the three experiments respectively. For each loiter circle, the h from the leakage source was varied just like Case-1.

B. Results

This section summarizes the key findings after the application of the two ML algorithms (ANN & RSVM) for leakage-detection on the data-sets generated by the three experiments for each of the two cases. The data values recorded in the SD card for each loiter circle during each experiment exhibited significant fluctuations. Hence, two leakage-detection values, viz., (i) the maximum value, and (ii) the minimum value were taken for each loiter circle for a specific h. Data-sets containing the maximum detection value were named as "Test-I", while the minimum detection value scenarios were labeled as "Test-II". Hence, for n loiter circles in an experiment, $2n$ values were available for training and testing of each algorithm.

The large variability created by using the maximum and minimum values per loiter circle can better compare the estimation performance efficacy of both algorithms. For both Test-I and Test-II value sets, 80% were used for algorithm training and remaining 20% for testing. The resulting graphs are presented in an experiment-wise fashion, comparing the detection efficacy for various heights. The red circles denote the experimental values, whereas the blue circles indicate estimations of the respective algorithms.

1) Experiment-1: The loiter radius remained constant at 0.5m for this entire experiment, while the leakage pressure was ∼1 PSI for Case-1 and ∼2 PSI for Case-2. The h increased from one meter upwards until the sensor failed to detect any gas concentrations. Further, both cases incorporated crosswinds at speeds around 2∼4m/s. Figures (6a) - (6d) depicts the estimation performance of the ANN in both cases, while Figures (6e) - (6h) illustrates the same for RSVM.

In Case-1 that models minor gas leaks at one PSI and the loitering radius at 0.5m; the gas-sensor (TGS2611) detected gas-leakage from a height of one meter upwards. The maximum detection height recorded for Experiment-1 was around eight meters. Though the gas-leakage detection values (PPM) were highly fluctuating, the general trend indicated that an increase in h tends to reduce the detection concentration values.

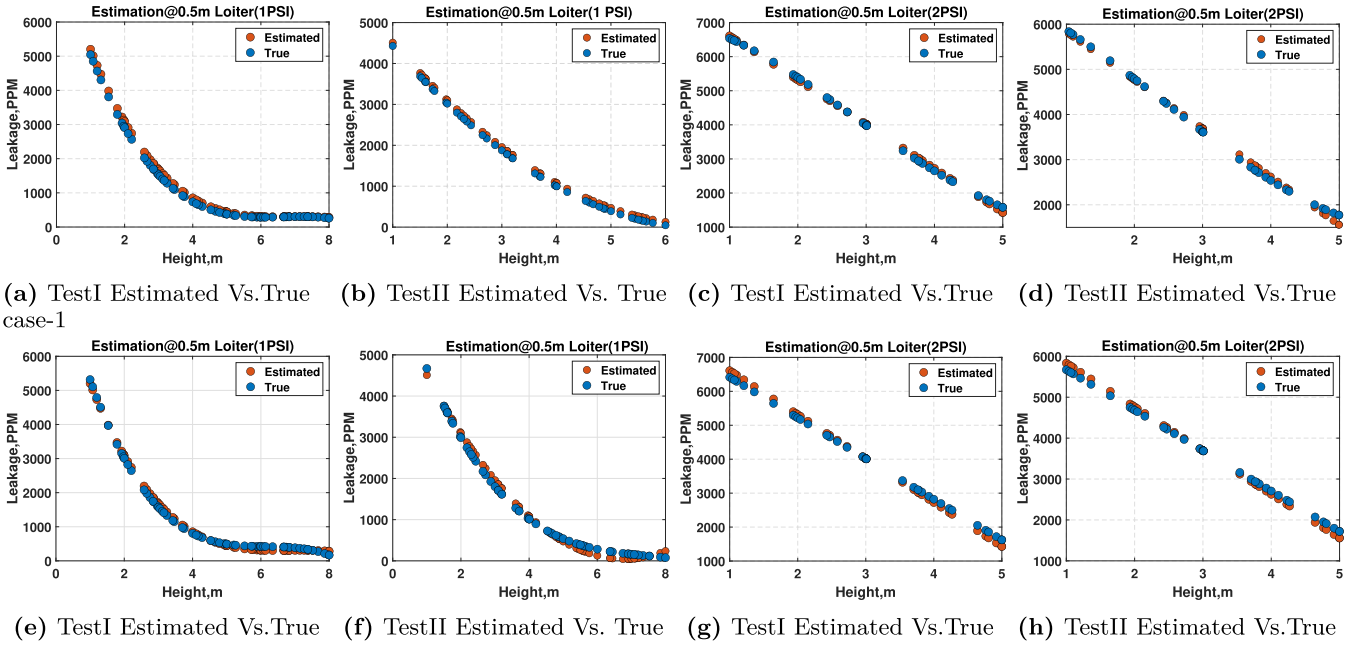
Figures (6a) and (6b) represent the ANN performance for maximum (Test-I) and minimum (Test-II) values of the Case-1 respectively, while Figures (6c) and (6d) are the same for ANN in Case-2 for Test-I and Test-II values respectively. It can be seen that in Case-1, the reductions in detection concentrations (PPM) by ANN follow an exponential distribution (Figures (6a) & (6b)), while for Case-2 (Figures (6c) & (6d)), it nearly follows a straight line with a negative slope. Similarly,

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TABLE I
COMPARISON OF ESTIMATION AND DETECTION PERFORMANCE OF ANN AND RSVM

Cases	Exp-1:(RMSE & R^2)	Time(s)	Exp-2:(RMSE & R^2)	Time(s)	Exp-3:(RMSE & R^2)	Time(s)
Case-1	[25.12, 88.18] (0.99, 0.985) [18.48, 122.96] (0.99, 0.99)	**{0.279,0.564}**	[21.87, 36.613] (0.99, 0.99) [27.65, 39.083] (0.99, 0.98)	**{0.272,0.552}**	[26.32, 45.78] (0.99, 0.97) [26.75, 38.08] (0.99, 0.995)	**{0.279,0.562}**
Case-2	[23.17, 121.21] (0.99,0.996) [24.63, 101.37] (0.99, 0.99)	**{0.278,0.560}**	[32.89, 48.17] (0.99, 0.99) [21.45, 112.54] (0.99, 0.99)	**{0.273,0.567}**	[25.32, 28.259] (0.99, 0.98) [19.62, 25.549] (0.99, 0.99)	**{0.272,0.557}**

[] contains the RMSE values , () provides R^2 values & {}Time(sec) in the order of [$RMSE_{ANN}$, $RMSE_{RSVM}$] & ($R^2{}_{ANN}$, $R^2{}_{RSVM}$) & {$Time(sec)_{ANN}$,$Time(sec)_{RSVM}$}



(a) TestI Estimated Vs.True case-1 (b) TestII Estimated Vs. True (c) TestI Estimated Vs.True (d) TestII Estimated Vs.True

(e) TestI Estimated Vs.True (f) TestII Estimated Vs. True (g) TestI Estimated Vs.True (h) TestII Estimated Vs.True

Fig. 6. Experiment 1: ANN performance-(a),(b),(c),(d) and RSVM performance-(e),(f),(g),(h).

the RSVM performance for both cases (Figures (6e) - (6h)) exhibit similar patterns as that of ANN. This clearly supports our hypothesis that gas-leakage detection is inversely proposal to the altitude (h) of the UAV.

The root mean square error (RMSE) and R-square (R^2) values of the ANN for Test-I & Test-II of the Experiment-1 for low pressure leakage scenario is provided in square parenthesis of cell labeled "Case-1" & "Exp-1" of the Table I, while the same for RSVM is in the circular parenthesis in the same cell. Similarly, the cell labeled "Case-2" & "Exp-1" presents results for moderate leakage scenario in the same fashion. In all situations of Experiment-1, the RSME values of ANN were within 15-28% of the corresponding values of the RSVM; suggesting lesser estimation error. Similarly, R^2 values of the ANN was either equal or better than those values reported by RSVM indicating better fit. Thus, the overall performance of the ANN is better than the RSVM in both cases for Experiment-1.

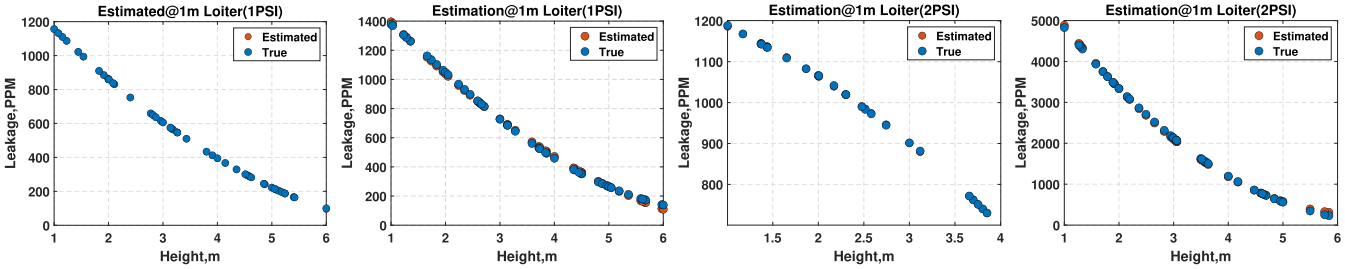
2) Experiment-2: Increasing the loiter radius to one meter was the major change in this experiment with that of the previous one. The maximum gas-leak detection height recorded for Experiment-2 was around six meters.



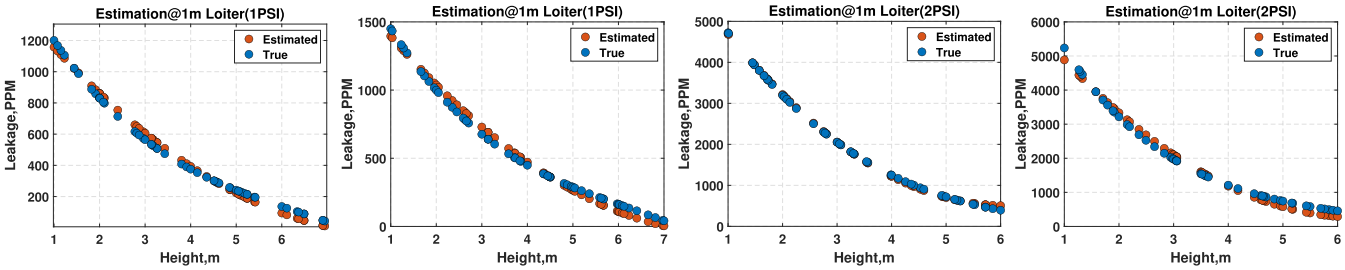
Fig. 7. View of ongoing experiment.

Figures (8a) - (8d) shows the performance of the ANN in both cases, while Figures (8e) - (8h) do the same for the RSVM. The RSME and R^2 values of both algorithms for Experiment-2 are presented in the same way as that of the

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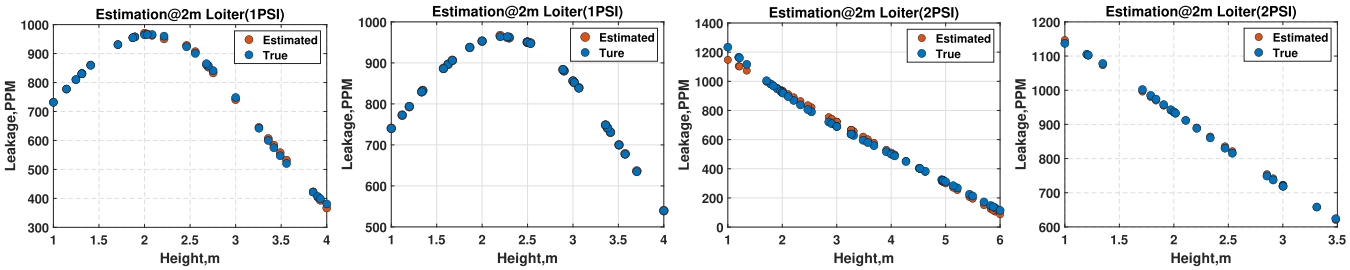


(a) TestI Estimated Vs.True case-1 (b) TestII Estimated Vs. True (c) TestI Estimated Vs.True (d) TestII Estimated Vs.True

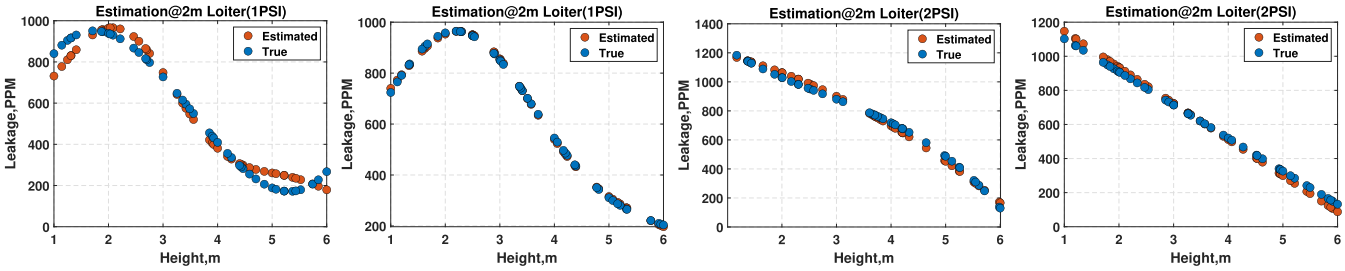


(e) TestI Estimated Vs.True (f) TestII Estimated Vs. True (g) TestI Estimated Vs.True (h) TestII Estimated Vs.True

Fig. 8. Experiment-2: ANN performance-(a),(b),(c),(d) and RSVM performance-(e),(f),(g),(h).



(a) TestI Estimated Vs.True case-1 (b) TestII Estimated Vs. True (c) TestI Estimated Vs.True (d) TestII Estimated Vs.True



(e) TestI Estimated Vs.True (f) TestII Estimated Vs. True (g) TestI Estimated Vs.True (h) TestII Estimated Vs.True

Fig. 9. Experiment-3: ANN performance-(a),(b),(c),(d) and RSVM performance-(e),(f),(g),(h).

previous experiment in cells titled "Case-1" & "Exp-2", and "Case-2 & "Exp-2" of Table I.

The ANN performance in the maximum (Test-I) and minimum (Test-II) values for Case-1 are illustrated in Figures (8a) and (8b) respectively, while Figures (8c) and (8d) do the same in Case-2 for ANN. Similarly, Figures (8e) and (8f) depict the RSVM estimation performance for Test-I and Test-II values for Case-1 respectively, and Figures (8g) and (8h) perform the same for Case-2 respectively. All these graphs indicate the reducing trend of the detection concentrations with respect to increasing values of h. Also, the estimated values by both algorithms fall closer to the measured values while using the maximum leakage detection values (Test-I).

The RMSE values for the ANN in both cases for Experiment-2 falls within 19-59% of those values reported by the RSVM, indicating better estimation (lower error). Similarly, the R^2 values reported by the ANN were at least the same or better than those values reported by RSVM, suggesting better fit to the measured data. This alludes to the superiority of the ANN over RSVM in estimating gas-leakage pressures reported by the UAV payload.

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3) Experiment-3: The third experiment differed from the previous experiments with the loitering radius increased to two meters, making the UAV farthest from the source of the leak. Crosswinds around 2∼4m/s were also part of this experiment as in the previous experiments. The more difficult detection conditions resulted in the detection height to fluctuate within four to six meters; indicating a reduction in the detection efficacy. The gas-leakage estimation performance of the ANN for Experiment-3 is illustrated in Figures (9a) - (9d), while Figures (9e) - (9h) do the same for the RSVM. Cells titled "Case-1" & "Exp-3", and "Case-2 & "Exp-3" of Table I summarizes the RMSE and R^2 values for both algorithms in both cases.

Unlike both Experiments 1 & 2, the leakage data behaviour in certain cases (Figures 9a, 9b, 9e, 9f) exhibits an an initial increasing trend followed by a sharp decreasing trend. All other cases exhibit the general pattern of reduction in detection concentration (PPM) with respect to any increase in the h. ANN continued to exhibit its superiority over RSVM in both RMSE and R^2 values, but unlike other two experiments, the RSME values of ANN were within 56-89% of this values reported by RSVM. In all cases, the ANN reported better R^2 values than RSVM.

V. Conclusion

This research compared the performance of two machine learning based estimation algorithms, viz., (i) artificial neural network (ANN), and (ii) reduced support vector machine (RSVM) using natural gas leakage detection data created from three experiments using a sensor based payload mounter on a multi-copter UAV. Both algorithms were compared in terms of accuracy, computational time, and goodness of fit of the regression. Results indicate that both methods are viable alternatives to identify and estimate gas leakage concentrations under varying and realistic operational conditions.

However, the neural network based regression performed better and faster in comparison with the support vector regression as summarised in Table I. It should also be noted that the accuracy and fit of support vector regression depends heavily on the kernel function. By replacing the simpler gas-sensor with better alternative like remote methane leak detector (RMLD) may enhance the detection performance of the multi-copter UAV. Thus, integrating the RMLD sensor with infra-red (IR) camera as payload of the UAV instead of the current payload may help in better and accurate detection of the amount of gas concentration around a designated area. With better data, the performance of the estimation algorithms could also be improved significantly.

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